

ARYAN SINGH

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EDUCATION

Vellore Institute of Technology **2026**
Bachelor of Technology in **Computer Science** (CGPA: **8.46**)

WORK EXPERIENCE

Beans.AI May 2025 – July 2025
Artificial Intelligence Intern

- Built a **RAG** (Retrieval-Augmented Generation) pipeline with **LangChain** and **ChromaDB** for driver access queries.
- Implemented semantic similarity search over vector embeddings of delivery notes, gate manuals and FAQs.
- Tuned prompt templates across open-source **LLMs** to cut hallucinations and keep driver answers short.

Beans.AI Jan 2025 – March 2025
Artificial Intelligence Intern

- Developed an **OCR** script in **Python** with **Tesseract** and **EasyOCR** to read unit numbers from driver photos.
- Applied **OpenCV** preprocessing (grayscale, thresholding, denoising) to improve accuracy in low light.
- Validated extracted text against internal geocoded databases to verify entrance and apartment locations.

PROJECTS

Fashion Mirror Studio — *Google Gemini, FastAPI, React 18, TypeScript*

- Architected a **two-stage generative pipeline** (Hero Anchor to Pose Fan-Out) on **Google Gemini Vision**.
- Built a **9-block prompt engine** and **dHash** reference selection behind a FastAPI/React workbench.

Sleep-HiT: Multi-modal Sleep Apnea Detection — *PyTorch, AST, Streamlit*

- Built a **hierarchical cross-modal Transformer** fusing tracheal audio (**AST**) with lifestyle data.
- Gated the 10–30s airway-to-SpO2 delay, reaching **89.1% F1** under LOOSO on the UCD dataset.

Delhi Transit & Urban Route Optimization — *Python, Transformers, PostGIS, GTFS*

- Designed a **Transformer** pipeline on Delhi bus network data to forecast route demand (R^2 **0.97**).
- Processed spatial-temporal **GTFS** data with **PostGIS** to surface underserved urban corridors.

TECHNICAL SKILLS

AI & Machine Learning: RAG, LangChain, LangGraph, CrewAI, Prompt Engineering, XGBoost, Random Forests

Deep Learning & CV: PyTorch, TensorFlow, Transformers (ViT, AST), CNNs (YOLOv8), OpenCV, Tesseract, EasyOCR

Data & GIS: ChromaDB, Pinecone, PostgreSQL, PostGIS, GeoPandas, Folium, SQL, Data Visualization

Cloud & Systems: AWS (EC2, S3, Lambda, SageMaker), Oracle Cloud, Azure, Docker, FastAPI, Streamlit

Languages & Tools: Python, Java, Go, C++, TypeScript, Git, Weights & Biases (W&B)

CERTIFICATIONS

Oracle Cloud Infrastructure (2025) – Certified Generative AI Professional | View credential